

Lab 12 JQuery

jQuery is a lightweight, "write less, do more", **JavaScript library**.

How to add JQuery ?

- ➔ By including CDN link in the program.(
<https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js>)
Import it in head tag:

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js" integrity="sha256-  
/JqT3SQfawRcv/BIHPThkBs00EvtFFmqPF/lYI/Cxo=" crossorigin="anonymous"></script>  
</head>
```

- ➔ By downloading JQuery from jquery.com

Syntax:

\$(selector).action()

Breakdown:

‘\$’ ‘’: indicates the start of jQuery

‘(selector)’ : selector means the html tag on which action is to be performed.

Example: \$("body") , \$("h1"), ... [use dot operator as prefix for class and # operator as prefix for id of the html element which will study in coming parts of this document].

‘action()’ : the action that will be performed on the selector.

Example To display a msg on clicking h1 Tag we write:

```
$('#h1').click(()=>{document.write('H1 is clicked ')})
```

Here **.click()** is the **action** that calls arrow function()=>{ document.write(“”) } to be executed.

JQuery Selectors:

jQuery selectors allow you to select and manipulate HTML element(s).

You can select and manipulate HTML elements by their name, id, class, attribute etc.

Element selector

\$(‘elementTagName’)

Examples: \$("h1"), \$("p"), \$("input") etc

Id Selector

`$('#idName')`

Here # represents that the selector is ID.

```
<button id="btn">Click</button>
  <p id="myEle">Hello World</p>

  <script>
    $('#btn').click(function (){
      $('#myEle').hide();
    })
  </script>
```

Class Selector

`$('.className')`

className is prefixed by . which represents that the class is selected.

Example:

```
<button id="btn">Click</button>

<p class="myEle">Hello World</p>
<p class="myEle">How are you??</p>

<script>
  $('#btn').click(function (){
    $('.myEle').hide();
  })
</script>
```

There are many such selectors

Selector	Description
<code>\$("tagName")</code>	Selects element of that tag name.
<code>\$(".className")</code>	Dot . is used to select specific element with className
<code>\$("#idName")</code>	# is used to Select specific element with idName.
<code>\$("*")</code>	Selects all elements
<code>\$(this)</code>	Selects the current HTML element
<code>\$("p.intro")</code>	Selects all <p> elements with class intro
<code>\$("p:first")</code>	Selects the first <p> element

Selector	Description
<code>\$("ul li:first")</code>	Selects the first of the first
<code>\$("ul li:first-child")</code>	Selects the first of every
<code>\$("[href]")</code>	Selects all elements with an href attribute
<code>\$("[target='_blank']")</code>	Selects all <a> elements with target="_blank"
<code>\$("[target!='_blank']")</code>	Selects all <a> elements whose target is not _blank
<code>\$(":button")</code>	Selects all <button> elements and <input type="button"> elements
<code>\$("tr:even")</code>	Selects all even <tr> elements
<code>\$("tr:odd")</code>	Selects all odd <tr> elements

JQuery Events

An event represents the precise moment when something happens. Examples

- moving a mouse over an element
- selecting a radio button
- clicking on an element

Some of the commonly used events are: click(), keypress(), submit(), load(), dblclick(), keydown(), change(), resize(), mouseenter(), keyup(), focus(), scroll(), mouseleave(), blur(), unload() etc.

```
<button id="btn1">Append</button>
<button id="btn2">Prepend</button>
<ul>
  <li>a</li>
  <li>b</li>
  <li>c</li>
</ul>
<script>
  $(document).ready(function (){
    $('#btn1').click(function (){
      $('ul').append("<li>Hello World</li>");
    })
    $('#btn2').click(function (){
      $('ul').prepend("<li>Hello World</li>");
    })
    $('#btn1').dblclick(function (){
      $('ul').append("<li>Double clicked on btn1</li>");
    })
  })
</script>
```

jQuery Event	Syntax	Callback Parameter(s)	Purpose / When it Fires	Example Use
click()	<code>\$(selector).click(function(event){ })</code>	event <i>(optional)</i>	Fires when the user clicks an element (left mouse click).	Hide, show, append data, change text, fade elements
focus()	<code>\$(selector).focus(function(event){ })</code>	event <i>(optional)</i>	Fires when an input field gains focus (user clicks or tabs into it).	Highlight textbox with background color
blur()	<code>\$(selector).blur(function(event){ })</code>	event <i>(optional)</i>	Fires when an element loses focus.	Form validation, restore background color
mouseenter()	<code>\$(selector).mouseenter(function(event){ })</code>	event <i>(optional)</i>	Fires when mouse pointer enters an element.	Change paragraph background color
mousemove()	<code>\$(selector).mousemove(function(event){ })</code>	event <i>(required if coordinates needed)</i>	Fires whenever the mouse moves over an element.	Print mouse coordinates
submit()	<code>\$("#form").submit(function(event){ })</code>	event <i>(commonly used to prevent submission)</i>	Fires when a form is submitted.	Validate complete form before submission

Method	Purpose	Parameters
hide()	Hide element	.hide(speed, callback function)
show()	Show element	.show(speed, callback)
toggle()	Toggle hide/show	.toggle(speed, callback)
fadeOut()	Fade element out	.fadeOut(speed, callback)
fadeIn()	Fade element in	.fadeIn(speed, callback)
fadeToggle()	Toggle fading	.fadeToggle(speed, callback)
slideUp()	Hide with upward slide	.slideUp(speed, callback)
slideDown()	Show with downward slide	.slideDown(speed, callback)
slideToggle()	Toggle slide	.slideToggle(speed, callback)

DOM Manipulation

Method	Purpose	Parameters
.text()	Get or set plain text	text() or text("Hello")
.html()	Get or set HTML	html("Hello")
.val()	Get or set input value	val() or val("ABC")
.append()	Add content at end	HTML/Text/Element
.css()	Change CSS	Property & Value
.children()	Select direct children	None
.siblings()	Select siblings	Optional selector